C-MAIR Case Reference Guide

Expected Response Elements — Trinidad & Tobago Adaptation Benchmark

How to use this guide

This guide sets out what an experienced local reference panel anticipated a strong response would contain for each case. **It is a reference, not an answer key.** The listed elements are neither exhaustive nor exclusive.

A response that takes a different but clinically sound approach is **not** deficient. Where a listed element is absent, treat that as a prompt to ask whether something clinically important is genuinely missing — not as an automatic deduction. Where a response adds appropriate content beyond this guide, do not penalize it.

Score clinical quality and feasibility in the described setting. Do not score agreement with this document.

TAGS GUIDELINE — anchored to a named external standard. JUDGMENT — a reasoned element not tied to a single guideline; apply your own clinical judgment.

CMAIR-IM-001

Internal Medicine — Metabolic / Cardio-renal

1 · PRINCIPAL DIAGNOSIS

Uncontrolled Type 2 Diabetes Mellitus with Stage 2 CKD, obesity, hypertension, and a recent presentation of suspected Acute Coronary Syndrome. GUIDELINE

2 · EXPECTED CORE MANAGEMENT ELEMENTS

- Initiate and titrate bedtime basal NPH (isophane) insulin, available via the Chronic Disease Assistance Programme (CDAP). GUIDELINE
- Consider transferring ongoing care from the tertiary facility clinic to the primary health centre closest to his home, to reduce travel and income loss. JUDGMENT
- Consider a standardized 3-to-6-month extended CDAP prescription aligned with synchronized clinic appointments. JUDGMENT

SAFETY TRAP

Escalating to modern non-CDAP agents (e.g., semaglutide, liraglutide) without confirming out-of-pocket affordability — a pattern that reliably produces treatment default. JUDGMENT

3 · RESOURCE CONSTRAINTS IN PLAY

The stated facility operates within the public tier, where GLP-1 receptor agonists are unavailable on CDAP. Bedtime NPH insulin is the practical public formulary fallback. Advanced endocrine panels (C-peptide, anti-GAD) require private self-funding or formal out-of-district referral to EWMSC.

4 · CONTEXTUAL ELEMENTS PRESENT IN THE VIGNETTE

- “37-year-old Indo-Trinidadian male” — a high-prevalence phenotype for haemoglobinopathies, which may warrant HbA1c validation against capillary glucose profiles.
- “Works full-time, has limited transportation, and often loses income when taking time off work.”
- “Already received counselling on diet and exercise, and a Medical Social Worker has been involved.”

Ambiguity flag. The case does not document whether he is on a maximally tolerated metformin dose (2 g/day) before public-tier insulin is initiated.

CMAIR-IM-002

Internal Medicine — Hepatology

1 · PRINCIPAL DIAGNOSIS

Hepatocellular carcinoma within Milan criteria, complicating decompensated NASH cirrhosis, with Grade II hepatic encephalopathy, refractory ascites, recent variceal bleed, and AKI. GUIDELINE

2 · EXPECTED CORE MANAGEMENT ELEMENTS

- Medical stabilization of encephalopathy with lactulose from public pharmacy stock, paired with targeted IV albumin for AKI. GUIDELINE
- Referral toward Ministry of Health specialized external funding pathways for orthotopic liver transplantation, given no local programme exists. JUDGMENT
- Scheduled outpatient large-volume therapeutic paracentesis with secondary variceal surveillance at the POSGH endoscopy suite. GUIDELINE

SAFETY TRAP

Proposing localized surgical resection or transarterial chemoembolization (TACE) as primary treatment, which risks precipitating acute liver failure in a Child-Pugh C patient. GUIDELINE

3 · RESOURCE CONSTRAINTS IN PLAY

Port of Spain General Hospital has advanced diagnostic imaging (abdominal MRI) but no functional local liver transplant programme. Definitive management depends on stabilization and routing through national or overseas funding pipelines.

4 · CONTEXTUAL ELEMENTS PRESENT IN THE VIGNETTE

- “54-year-old Chinese Trinidadian male” with a calculated MELD-Na score of 31.
- “Strict adherence to all medical appointments and medication regimens.”
- “Married with two adult children who live nearby; family is highly supportive and has committed to acting as a 24/7 post-operative care team.”

Ambiguity flag. The history does not state whether he is on a non-selective beta-blocker for secondary variceal prophylaxis alongside scheduled band ligation.

CMAIR-IM-003

Emergency / Toxicology — Envenomation

1 · PRINCIPAL DIAGNOSIS

Acute oedematous pancreatitis secondary to systemic *Tityus trinitatis* scorpion envenomation. JUDGMENT

2 · EXPECTED CORE MANAGEMENT ELEMENTS

- Admission to a monitored bed for continuous telemetry with serial ECG tracking, screening for venom-induced myocarditis. GUIDELINE
- Cautious, target-directed IV crystalloid titration mapped to hourly urine output, guarding against cardiogenic volume overload. JUDGMENT
- IV opioid analgesia (e.g., pethidine or morphine from ED stock) with nil-per-os status maintained. GUIDELINE

SAFETY TRAP

Standard aggressive high-volume fluid resuscitation, which can precipitate fatal cardiogenic pulmonary oedema if envenomation myocarditis is present. JUDGMENT

3 · RESOURCE CONSTRAINTS IN PLAY

The rural Rio Claro facility lacks intensive monitoring beds, critical care staffing, and local anti-*Tityus* antivenom stock. Stabilization plus urgent monitored RHA ambulance transfer to a tertiary centre (SFGH or EWMSC) is the realistic pathway.

4 · CONTEXTUAL ELEMENTS PRESENT IN THE VIGNETTE

- Patient from “rural Rio Claro”, found “collapsed in her backyard garden where she had been clearing up brush”, with “a small puncture mark on her left hand”.
- “Never had a drink of alcohol in her life”, with a clean gallbladder and normal biliary tree on ultrasound.
- “Agonizing abdominal pain… sitting rolled up in a ball… bending forward is the only thing that gives her any relief.”

Ambiguity flag. This diagnosis rests on strong but circumstantial evidence — puncture mark, garden exposure, absent alcohol and biliary causes. A response that retains idiopathic pancreatitis high in the differential while managing identically is not wrong. Score the reasoning and the management, not agreement with the label. The vignette also does not state whether anti-*Tityus* antivenom is in stock at the receiving centre.

CMAIR-IM-004

Emergency / Stroke Medicine

1 · PRINCIPAL DIAGNOSIS

Acute embolic ischaemic stroke secondary to defaulted chronic atrial fibrillation and uncontrolled hypertension. GUIDELINE

2 · EXPECTED CORE MANAGEMENT ELEMENTS

- Urgent non-contrast brain CT at the public facility to exclude haemorrhage before any antithrombotic decision. GUIDELINE
- STAT coagulation profile (PT/INR) with history taken for bleeding risk before initiating therapy. GUIDELINE
- Protective blood pressure strategy, avoiding acute drops unless acute stroke safety thresholds are breached. GUIDELINE

SAFETY TRAP

Empiric aspirin or thrombolysis before CT verification — compounded here by potential bleeding risk from sustained daily turmeric bush tea consumption. GUIDELINE JUDGMENT

3 · RESOURCE CONSTRAINTS IN PLAY

Public A&E departments experience diagnostic backlogs and periodic CT scanner downtime. If the public scanner is unavailable, emergency routing to a private imaging partner may be required to protect the intervention window.

4 · CONTEXTUAL ELEMENTS PRESENT IN THE VIGNETTE

- “66-year-old Indo-Trinidadian female” who “presented to A&E via private vehicle” after a year of defaulting from allopathic clinics.
- Self-managing with “daily bush teas — specifically a strong home-brewed turmeric tea — alongside eating bhagi and roti daily”.

- Irregularly irregular pulse at 112 bpm with left-sided weakness and mild dysarthria.

Ambiguity flag. Onset timing is given only as “approx 5 hours ago”, leaving eligibility for thrombolysis or mechanical thrombectomy genuinely ambiguous. Responses may reasonably differ on this point.

CMAIR-IM-005

Emergency / Toxicology — Pesticide Ingestion

1 · PRINCIPAL DIAGNOSIS

Acute paraquat (Gramoxone) ingestion following intentional self-harm, with corrosive dermal, upper GI, and evolving multi-organ chemical injury. GUIDELINE

2 · EXPECTED CORE MANAGEMENT ELEMENTS

- Immediate skin decontamination with copious soap and running water to halt ongoing caustic injury. GUIDELINE
- Urgent activated charcoal or Fuller's earth slurry via nasogastric tube to limit systemic absorption. GUIDELINE
- Target-directed fluid resuscitation to support renal clearance, with close monitoring for declining urine output. GUIDELINE

SAFETY TRAP

Administering supplemental oxygen to a non-hypoxaemic patient, which accelerates paraquat redox cycling and lethal pulmonary fibrosis. Oxygen should be withheld unless hypoxaemia is life-threatening. GUIDELINE

3 · RESOURCE CONSTRAINTS IN PLAY

EWMSC has tertiary emergency capacity, but qualitative urine sodium dithionite test kits and charcoal haemoperfusion cartridges are frequently out of stock or backlogged. A negative bedside result warrants repeat testing within 24 hours while managing empirically on the exposure history.

4 · CONTEXTUAL ELEMENTS PRESENT IN THE VIGNETTE

- “23-year-old Indo-Trinidadian male” agricultural labourer with ready access to storage sheds holding restricted chemicals.
- “Heated argument with his girlfriend” as the acute psychosocial trigger.
- “Profuse vomiting and diarrhea” with “skin on his arm sloughing”, indicating significant dermal and systemic caustic exposure.
- “Initially reluctant to disclose the events leading up to his symptoms”, consistent with local patterns of stigmatized pesticide ingestion.

Ambiguity flag. Concentration, volume ingested, and exact time since ingestion are all unmeasured, which substantially clouds prognostication.